

Detection, tracking, and identification of drones: an overview on Counter-UAS techniques, and open challenges

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Abstract—With the proliferation of Unmanned Aircraft Systems (UASs), a broad spectrum of legitimate, nuisance, and malicious applications has introduced new challenges for airspace management. A large number of UASs available on the market, and UASs can also be homemade (including being 3D printed), further complicating detection and classification efforts. Traditional traffic control and surveillance systems are facing disruption from UASs proliferation, which enable innovative use cases across various sectors, such as precision agriculture and infrastructure inspection, but also create opportunities for misuse, such as illegal surveillance and contraband delivery. In this evolving landscape, C-UAS technology has emerged as a crucial tool to mitigate the growing risks associated with unauthorized drone usage. To this end, it is critical to understand the capability, limitations and challenges associated to C-UAS research and development efforts related to detection, tracking, and identification of unauthorized UASs in public and critical infrastructures. Given the diverse range of sensors employed in C-UAS systems, such as radar, radiofrequency, electro-optical, and acoustic sensors, and the heterogeneous nature of UASs, this study emphasizes the importance of multi-sensor data fusion to achieve accurate DTI in real-time. The integration of different sensors is key to enhancing system reliability and addressing the complexities of modern airspace. Furthermore, new models and algorithms are necessary to improve target tracking capabilities for UASs. The study presents a taxonomy of sensors used in C-UAS and highlights the need for a comprehensive, integrated approach. It also delves into the challenges of implementing a multi sensor data fusion algorithm, providing a model for sensor fusion in DTI, along with an overview of the current literature and available commercial solutions. By adopting a holistic approach, this study aims to contribute to the advancement of robust, reliable C-UAS systems that can effectively manage the rising threat of unauthorized UASs.

Index Terms—Counter-Unmanned Aircraft System (C-UAS), Unmanned Aerial Vehicle (UAV), Unmanned Aircraft System (UAS), drones, critical infrastructures, counter-terrorism, Space-Air-Ground Integrated Network (SAGIN)



Fig. 1: The envisioned 6G Internet integrates Non-Terrestrial Networks (NTNs), satellites, HAPs, and UASs, with TNs on land and at sea.

I. INTRODUCTION

DRONES¹ are democratizing access to airspace through cost-effective and user-friendly platforms, disrupting aviation and related industries. Their transformative applications span urban air mobility, logistics, pharmaceutical delivery, emergency reconnaissance, precision agriculture, and geospatial surveying [1]. In telecommunications, UASs have emerged as pivotal tools for enhancing connectivity in congested urban and smart city environments, where drones provide seamless handover between terrestrial networks (TNs) and high-altitude entities, e.g., satellites and High-Altitude Platforms (HAPs), as illustrated in Fig. 1. To this end, UASs are considered as key enablers for Sixth Generation (6G) networks and integrated Terrestrial/Non-Terrestrial Network (T/NTN) architectures.

However, the proliferation of drones has also raised significant security and safety concerns. Malicious actors exploit

¹They are also known as Unmanned Aerial Vehicles (UAVs) or Remotely Piloted Aircraft Systems (RPASs). In regulation, the term commonly used is Unmanned Aircraft Systems (UASs). In line with EU and USA regulations, we use UAS to refer to the whole system, including pilot, operator, data-links, and Unmanned Aircraft (UA) [2]–[4].

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their accessibility and ease of use for illicit activities, e.g. smuggling, terrorism, and unauthorised access to sensitive sites [5]. Recent conflicts, contraband, drug trafficking, and especially reconnaissance near critical infrastructure, underscore the urgency of mitigating non-cooperative and non-compliant drones. These threats can be broadly distinguished in (i) criminal, (ii) unlawful, and (iii) accidental.

Regulatory frameworks are evolving [3], [6], [7] to ensure safe operations, particularly for Beyond Visual Line of Sight (BVLoS) flights, which enable fully autonomous UAS deployment at scale. While Unmanned Traffic Management (UTM) systems oversee these operations and airspace access, the lack of direct human oversight leads to vulnerabilities. Adversaries could exploit BVLoS capabilities to target public spaces, infrastructure, or crowds [8], necessitating robust Counter-Unmanned Aircraft System (C-UAS) to safeguard these zones.

The main purpose of C-UAS solutions is the detection, tracking, and identification (DTI) of unauthorised drones within a designated “scan area” through multi-modal sensors, i.e., Radio Detection and Ranging (RADAR), radiofrequency (RF) passive receivers, and acoustic detectors. While civilian systems would focus on detecting in time the unlawful use of drones, military implementations extend to neutralisation and forensic analysis [9].

While this field is nascent for drones, insights from aviation, Internet of Things (IoT), Earth imaging, automotive, and biomedical sensing offer transferable solutions for its challenges [10], [11].

A core challenge lies in processing the heterogeneous data streams generated by C-UAS sensors, which affect the quality and accuracy of DTI [10]. These data sets vary for being (i) multi-modal, i.e., encompassing different sensor types, mainly optical, thermal, RADAR, and RF; (ii) high-dimensional, e.g., hyper-spectral sensors produce complex feature spaces; (iii) multi-scale, i.e., varied spatial resolutions and sampling rates; and (iv) mixed structure data, e.g., vector or raster, and are affected by non-linear correlations.

This complexity introduces networking bottlenecks, synchronisation issues, and computational demands, exacerbated by wireless sensor constraints and environmental interference, e.g., fog and sunlight. To meet real-time requirements, C-UAS architectures require (i) adaptive machine learning pipelines resilient to noisy, imbalanced, or incomplete data; (ii) efficient feature selection and tuning for rapid ingestion of continuous data streams; and (iii) distributed and federated systems, e.g., the air traffic management surveillance tracker and server (ARTAS) network used in aviation [12], to enable collaborative learning and early threat detection across the airspace [10].

In this context, this paper offers recent advancements and open challenges in C-UAS development, proposing an updated sensor taxonomy and emphasising the necessity of multi-sensor fusion frameworks for reliable DTI. A preliminary evaluation demonstrates the efficacy of multi-modal C-UAS architectures, while outlining current limitations and future research directions to develop the field.

II. RELATED WORK

Recent literature focuses on adapting traditional aviation principles for DTI of manned aircraft [13] to address the unique vulnerabilities and intrusion strategies of UASs [14], [15]. These advancements are driving a surge in research and commercialisation efforts, as evidenced by a growing trend of innovative C-UAS solutions [11], [16]–[21].

Governments are establishing guidelines to regulate C-UAS deployment in sensitive areas, with growing emphasis on integration into future Air Mobility ecosystems [22]. For instance, the Joint Research Centre (JRC) of the European Commission is spurring C-UAS innovation through initiatives such as its DRONE project, which has produced handbooks on hardening critical infrastructure and protect public spaces against drone threats [23]–[25]. Additionally, the JRC operates a framework contract for C-UAS services to protect European Commission facilities across Belgium, alongside plans to establish a *Living Lab*² for real-world testing.

Complementing these efforts, the European Union (EU)-funded COURAGEOUS project [26], launched in 2021, aims to standardise testing methodologies for evaluating C-UAS performance in DTI of rogue drones. By defining user-defined scenarios spanning diverse operational contexts, the project enables objective, quantitative comparisons of C-UAS solutions, advancing the protection of low-altitude airspace. Building on this, [20] analyses emerging trends in C-UAS design, highlighting a preference for RADAR, RF and visible-light cameras over acoustic, Light Detection and Ranging (LiDAR), and infra-red (IR) sensors. While RADAR and RF are often deployed independently, hybrid architectures combining multiple types of sensors, i.e. multi-modal, are gaining traction.

Cooperative C-UAS frameworks, i.e., DTI systems based on a network of multiple C-UAS deployments, provide enhanced surveillance and UAS patrolling [27]. Such systems demand resilient wireless networks capable of resisting interference and cyberattacks, as detailed in [15], [28], [29], particularly for cellular and Wi-Fi-based systems [30].

Deep learning (DL) is increasingly integral in DTI frameworks, with multi-layer perceptron neural networks proving effective for RADAR-based classification tasks [31], [32]. However, DL models face hurdles when fed with noisy data and put under real-time processing constraints, necessitating optimised feature selection and careful pipeline design [33]–[35].

The design and integration of sensors for C-UAS systems is pivotal to their efficacy in DTI operations. This synthesis of current advancements, from European projects to sensor fusion strategies, underscores the need for continued innovation to address evolving UAS threats. By mapping ongoing challenges

²Living labs, such as the one in development in Geel (Belgium), are innovative, user-centric ecosystems designed to foster co-creation, innovation, and start-up growth. Living labs at JRC.

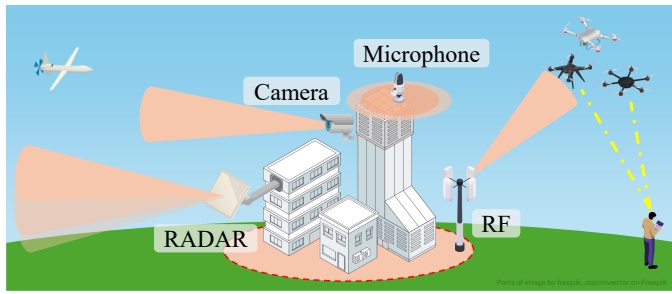


Fig. 2: Overview of a C-UAS deployment protecting a sensitive area, delineated by the dashed red line. Key components include: a RADAR system, an RF antenna for radio signal interception, a camera for visual/thermal detection, and an acoustic microphone array for drone noise analysis.

and research trajectories, this analysis provides a roadmap for advancing C-UAS technologies toward robust, scalable solutions.

III. TAXONOMY AND CHALLENGES OF C-UAS SENSORS

Sensors can be classified in various ways [10]: (i) based on whether they detect electromagnetic or mechanical waves; (ii) based on the specific spectral bands they operate within; (iii) based on their signal analysis methodology, distinguishing between active sensors, which generate and analyse their own signals, and passive sensors, which detect ambient signals; (iv) depending on their operational conditions; and (v) based on their deployment platforms. The set of sensors employed in a C-UAS system can be broadly illustrated as referenced in Fig. 2. Each combination of them introduces trade-offs between detection accuracy, latency, implementation cost, and computational resource demands. These factors directly impact data quality and the overall efficacy of the solution.

Sensors can be further classified by deployment platform [10]. These include fixed installations, such as ground-based towers, or mobile units on vehicles or personnel. Sensors can also be put on aerial platforms, such as collaborative UA, which enhance C-UAS capabilities by extending the DTI operational range. Their variable operational altitude reduces environmental interference and clutter, improving signal fidelity wrt. ground-based systems.

A. RADAR

RADARs exhibit diverse configurations, each offering trade-offs in performance, cost, and technological maturity. These systems operate by transmitting radio signals and then analysing backscattered signals from the scan area, including potential targets. Amplitude and phase shifts in reflected signals indicate target intensity and distance, while Doppler analysis estimates velocity. Recent studies highlight RADAR's growing role in C-UAS systems, with advancements in mmWave technology [36], micro-Doppler signature analysis [37], and tailored methodologies for drone detection [38], [39].

A key strength of RADARs lies in their robustness across weather conditions, though heavy rain and higher operating

frequencies, especially in the mmWave bands, exacerbate signal attenuation [20]. Sensor selection hinges on wavelength coverage, typically spanning 1 mm (300 GHz) to 100 m (300 MHz) [10], aligning with the wide size range of UASs wingspans, i.e., 1 mm to 61 m. Market trends reveal that 45% of C-UAS deployments omit RADAR range specifications, focusing instead on DTI ranges between 3 to 5 km and 7.5 to 10 km [40].

Deploying RADARs in C-UAS requires careful consideration of spectrum regulations, power constraints, interference, blind spots, and spatial resolution [20]. Emerging techniques such as Automatic Target Recognition (ATR) [41] leverage DL to classify drones via echo signals, though challenges persist in dataset scalability, real-world validation, and explainability for trustworthy deployments.

Synthetic Aperture RADARs (SARs), the most widely adopted RADAR variant, excel in high-resolution imaging independent of time or weather. Their mobility enables enhanced aperture synthesis, particularly when mounted on circular or moving platforms [42]. Recent research integrates ATR with SARs [43]–[45], though real-time processing demands—often overlooked—necessitate lightweight neural architectures and efficient data compression.

RADAR technology is rapidly evolving through Artificial Intelligence (AI) integration, quantum advancements, networked systems, and high-resolution imaging, expanding applications in C-UAS, defense, autonomous vehicles, and medical diagnostics.

B. Radiofrequency sensors

A UA operates as part of a broader UAS, which includes additional aspects such as remote control, telemetry, and data streaming via electromagnetic signals to ground stations or pilots [11]. These systems rely on radio transmissions for connectivity, navigation, and Internet services, emitting detectable and recognisable electromagnetic signatures.

Software-Defined Radios (SDRs) are widely used to intercept UAS radio activity, enabling localization through different techniques, e.g., Direction of Arrival (DoA) Time Difference of Arrival (TDoA). Advanced signal processing algorithms achieve accurate direction finding with minimal hardware, starting from two antenna pairs [46]. Traditional RF analysis and Wi-Fi interception face challenges in open-field environments, particularly for wide areas, due to the need for high-gain antennas, a scalable system, or the employment of specialized networks [47]. Current research is further limited by datasets (e.g., [48]) focused on single-drone laboratory tests, hindering the development of scalable C-UAS solutions capable of DTI for swarms or discriminating legitimate traffic from adversarial spoofing.

Circumvention of Direct Remote Identification (DRI) also known as Remote ID [49] poses significant regulatory and privacy challenges [50]–[52]. Autonomous drones without radio emissions can be detected via bistatic RADAR setups, which analyse the Channel State Information (CSI) for interference caused by drone propellers [53]. Similarly, Global Positioning

System (GPS) reflectometry leverages satellite signals to detect UASs by analysing environmental reflections [10].

Full-duplex RF systems, capable of simultaneous transmission and reception, simplify deployments by reducing hardware requirements. Military trials in the 225 ~ 400 MHz band up to 100 W and Fifth Generation (5G) in-band tests at 2.4 GHz demonstrate their potential [54], [55].

DL models significantly improve DTI accuracy by extracting frequency-based features unique to specific drone models and flight configurations [56]. These models outperform conventional sensors in classifying UASs and filtering false signals.

C. Optical sensors

In this domain, we distinguish active sensors, which are predominantly LiDAR systems that allow precise ranging, from passive sensors, which are visible-light cameras, IR ones, multispectral sensors for discrete wavelengths, and hyperspectral ones for contiguous narrow bands [10]. These systems can be enhanced with Pan-Tilt-Zoom (PTZ) capabilities for dynamic surveillance in case of limited field of view [20].

While DL-enhanced computer vision has advanced electro-optical sensor detection and positioning capabilities [57], key limitations hinder their adoption in C-UAS, particularly on (i) environmental sensitivity, due to reliance on ambient light, thus the sensors are ineffective at night, vulnerability to adverse weather conditions or lens contamination without special equipment; (ii) hardware constraints, mainly due to fixed focal lengths imposed by the lens selection and detection capability limited by the sensor's resolution; (iii) localisation challenges, which require DL-enabled monocular set up [58] or hybrid Light Amplification by Stimulated Emission of Radiation (LASER)-ranging; (iv) PTZ limitations, due to restricted rotational speed and dynamic blind spots during zoom adjustments [20].

Short-wave IR (SWIR) imaging is emerging as a promising solution, especially for low-light and adverse weather conditions [27]. Additionally, few-shot learning techniques mitigate dataset scarcity by enabling drone classification with minimal training examples [57].

D. Acoustic sensors

Spatially distributed microphone arrays leverage the distinct rotor blade noise of UAs for DTI [11]. Localisation of UAs is done through methods such as TDoA, achieving high accuracy in controlled environments. However, performance degrades significantly if the sensors do not cover the entire scan area, especially in altitude, or in case of adverse weather phenomena. Furthermore, the detection is hindered when ambient noise levels overwhelm drone acoustic signatures.

Consequently, acoustic-based DTI, whether active or passive, is rarely prioritised for a stand-alone C-UAS set up [11]. Its utility lies primarily in hybrid sensor architectures, where acoustic data complements other modalities, e.g., RADAR and RF, to enhance their robustness.

IV. TOWARDS A RELIABLE, ROBUST, AND ACCURATE C-UAS: MULTI-SENSOR DATA FUSION

Due to the limitations of individual sensors, C-UAS solutions relying on a single modality face severe constraints in their reliability and accuracy [59]. A robust alternative involves multi-sensor architectures pair with data fusion algorithms, which integrate complementary sensor outputs to enhance DTI performance, even under incomplete or uncertain data conditions [60].

Data fusion operates at (i) feature level, which integrates raw or processed data from heterogeneous sensors; at (ii) decision level, which combines outputs from individual sensor classifiers; and at (iii) rule level, which applies pre-defined logic for sensor integration, trading-off adaptability and simplicity for abstraction [59], [61]. Key challenges include (i) sensor heterogeneity, i.e., the management of diverse data formats, resolutions, and sampling rates; (ii) spatio-temporal misalignment, as data streams may be asynchronous and spatially mismatched; and (iii) data complexity, as there is the need to process high-volume, multi-modal data with non-linear correlations [6], [35], [61].

Effective fusion requires synergies between sensors and their classifiers with the goal of maintaining DTI consistency, despite sensor failures, e.g., passive RF blind spots for autonomous UAs. Commercial systems often pair RADAR with optical cameras for independent tasks, or combine multiple cameras with passive RF to boost accuracy and identification. However, quantitative analyses for system optimisation remain scarce, with most research limited to qualitative reliability assessments [20].

Moreover, proprietary C-UAS solutions rarely disclose fusion methodologies, vaguely citing AI or classical statistical techniques. This opacity obstructs reproducibility and hinders improvements in multi-sensor integration, a process still hindered by technical complexities and the absence of standardised practices [20].

A. Preliminary Multi-Sensor Tracking Simulation

To demonstrate the potential of multi-sensor data fusion, we describe a simulation campaign to: (i) evaluate a RADAR target tracking scenario, (ii) analyse an RF-based approach, and (iii) explore how combining the two can enhance their effectiveness in DTI UAs. Data fusion is applied both at the sensor, to extract individual tracks, and across sensors, in order to create a system-level track for each detected target.

As illustrated in Fig. 3, the simulation of time T , discretised into K time slots, consists in a Cartesian space which hosts a RADAR $\mathbf{q}^R$ and a RF sensor $\mathbf{q}^F$ at coordinates $\mathbf{q}^R = \mathbf{q}_1^F = [0, 0, H]^T$. Moreover, $N - 1$ RF sensors are uniformly distributed at a radius of R^F around $\mathbf{q}_1^F$, denoted as $\mathbf{q}_i^F$ with $i = 2, \dots, N$. The notation and values adopted in the actual implementation are summarised in Table I. It is assumed that all the sensors are placed on platforms high enough to ignore ground clutter. Finally, U drones are simulated, having U-shaped trajectories expressed as waypoints denoted by $\mathbf{w}_{u,j}^U$, with $u = 1, \dots, U$, $j = 1, \dots, J$ representing

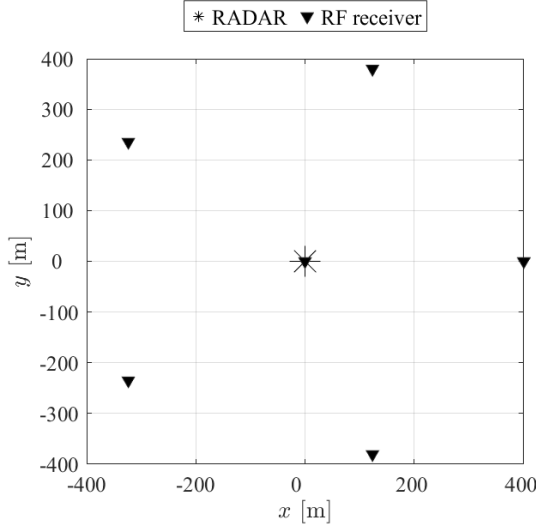


Fig. 3: Overview of the sensor set up in the simulation.

Parameter	Value	Parameter	Value
H	15 [m]	S^R	180 [deg/s]
$\mathbf{v}$	[6, 60] ^T [deg]	P^R	0.8
R^R	5 [km]	F^R	10^{-6}
T, K	50 [s, #]	V	6×10^{-8} [m ³]
R^F	400 [m]	M	20 [#]
β	10^{-4} [# / m ³]	G	20 [#]
T^B	5 [#]	S^N	10 [#]
T^D	1 [s]	V^U	6 [m/s]
a^U	6.3 [m/s ²]	f	5.8 [GHz]
B	80 [MHz]	U	1 [#]

TABLE I: Parameter settings.

the u -th UA and its j -th waypoint, respectively. It is worth remarking that the motion characteristics, indicated in Table I, are taken from commercial drones, i.e., DJI Matrice 300 RTK and Sony Airpeak S1, and expressed as V^U and a^U for the maximum speed and acceleration, respectively. Furthermore, the trajectory properties are summarised in Table II for each scenario.

1) *RADAR-only simulation*: assume the RADAR is a rotator with a scan rate S^R , a field of view $\mathbf{v} = [\theta, \varphi]^T$, with θ and φ the azimuth and elevation angle, respectively, characterised by a probability of detection P^R , a false alarm rate F^R , a range R^R , and a sensor bin volume V . The parameter values are based on actual commercial C-UAS implementation [62], using reasonable values if not specified by public data sheets.

Scenario #	1	2
UA height [m]	30	100
UA Waypoints	[-50, -50]	[-200, 500]
	[40, -50]	[-110, 500]
	[45, -44]	[-104, 506]
	[45, 45]	[-104, 596]
	[40, 51]	[-110, 601]
	[-62, 51]	[-212, 601]

TABLE II: Configured height and UA trajectories for each scenario. Waypoints are rounded to the nearest unit for brevity.

The Joint Probabilistic Data Association (JPDA) multi object tracker with Interacting Multiple Model (IMM) filter is implemented in MATLAB and configured with a maximum number of tracks M and a spatial density of new targets per unit volume β . Finally, the tracker has been configured with an absolute time tolerance between detections T^D .

The RADAR position and the UAs trajectories are shown in Fig. 4a. At a height of 30 m, the UA is not visible by the RADAR and hence no detection is provided. Conversely, the drone at 100 m of altitude is detected and shown in Fig. 4b. Based on these detections, the track is then reconstructed with JPDA and shown in Fig. 4c.

2) *RF-only simulation*: assume the RF sensors are passive receivers synchronised and tuned to intercept communications at frequency f with a bandwidth B . The UAs send a video stream to a remote pilot at different sub-carriers with sufficient spacing to not interfere with each other. Furthermore, the communication takes place with isotropic antennas with unit gain. In order to locate targets, the TDoA of these signals are evaluated to generate detections and the track is then reconstructed using Global Nearest Neighbour (GNN). Finally, the sensors' height are set with a uniform random variable between $[H, 2H]$.

Differently from the RADAR, both drones at 30 and 100 m of altitude are detected, although with less accurate tracks. An illustration of these two scenarios is available in Fig. 5. The tracks shown in Fig. 5 have already been processed for track reconstruction. However, we highlight that the JPDA algorithm is not effective for track reconstruction when applied to RF sensor data. This limitation suggests the need for further investigation to enhance the accuracy and reliability of the tracking process.

3) *Combined approach*: a combination of these two separate systems, i.e., RADAR and RF-based detection, leads to the fusion of the target tracks into a system track, which would enrich the amount of information recovered from a UA [63], [64]. For instance, if a drone is detected and tracked by the RADAR, the RF information provides further insights on the fact that the UA may not be autonomous, but it is remotely controlled and/or supervised. Moreover, using both sensors reduces the likelihood of missing a drone detection and decreases the probability of false alarms (e.g., a bird as a drone).

In order to create system tracks, the longest tracks are taken from each sensor. Consequently, the Dynamic Time Warping (DTW) algorithm is employed after K time slots, as introduced in [65]. Once the distance matrix is calculated, the system tracks are identified by coupling together the RADAR and RF tracks with the minimum distance between them.

B. Current limitation and future work

The TDoA scenario, presented in Sec. IV-A2, requires the maintenance of a set of coherent passive receivers placed far apart, in the hundreds of metres, which may not be practical. RF sensors can also be employed to identify the command and control communication protocol being used by the drone,

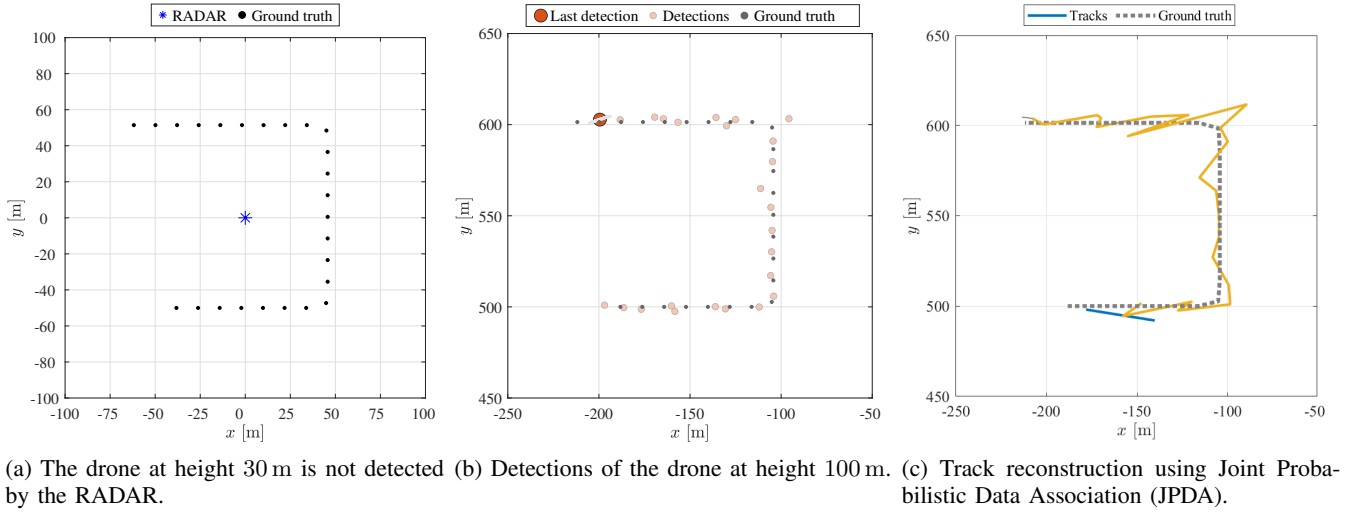


Fig. 4: Drone detection and track reconstruction done by the RADAR.

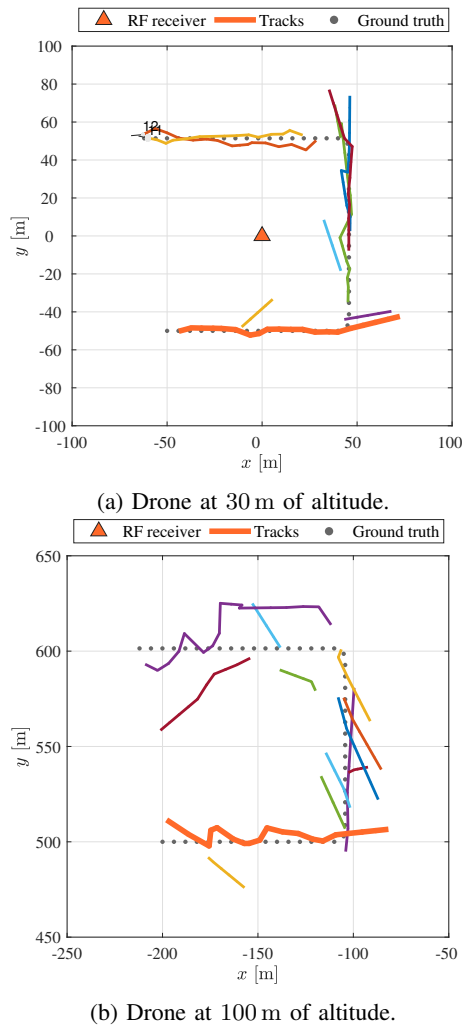


Fig. 5: Tracking done by the RF receivers.

Sensor type	Track fusion algorithm	Target dynamics filter
RADAR	JPDA [66]	IMM [67, Sec. 15.2.2.1]
RF	GNN [13]	Constant-velocity extended Kalman filter [67, Sec. 15.2.2]

TABLE III: Overview of the different tracking methodology used across the simulation campaign.

as well as to provide angle-of-arrival of the signal without having a coherent set up. Such information may be critical to study inter-sensor strategies for optimal coordination, as the RADAR, as well as cameras, may be notified of possible targets coming from a certain direction.

Furthermore, the combined approach mentioned in Sec. IV-A3 leads to the creation and update of system tracks every K time slots. Furthermore, the DTW implementation assumes that both RADAR and RF sensors are able to detect the drones. These two aspects hinder the applicability of such solution in real-world and real-time applications. A more sophisticated approach is required in order to fuse tracks together, even in case of incomplete data from the sensors.

V. CONCLUSION

This contribution highlights the recent advancements in terms of DTI of UASs by using multiples types of sensors, i.e., RADARs, RF, electro-optical, and acoustic, for C-UAS implementations. The most relevant and recent works are reported, together with key reviews to be able to analyse the ongoing open challenges in research. Furthermore, additional details are reported for the current trends in real-world implementations of C-UAS made by the growing industry of drone monitoring. Further work should focus on providing deeper insights on multi-sensor data fusion, which is a growing trend to DTI multiple UASs by employing multiple sensor types together, in order to increase the overall accuracy and robustness of C-UAS solutions. Finally, a preliminary multi-

sensor tracking simulation has been modelled and discussed, highlighting limitations and future works.

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